

MEC 422/ MEC 422: INDUSTRIAL ENGINEERING

Introduction

The phrase “Industrial Engineering” can only be better understood when we look at the two words separately.

Engineering: This is a profession that harmonizes the mathematical and natural science knowledge gained through study, experience, and practice. This same is judgmentally applied for development of ways to use materials and forces of nature effectively and economically for the benefit of mankind.

Industry: This is an economic activity concerned with the processing of raw materials and manufacture of goods in factories using necessary techniques.

Industrial Engineering: This entails intuitive design, improvement, and installation of integrated systems of people, material, energy and suitable technology with much emphasis on production, manufacturing, human factors area, and operations research. This is achieved keeping in mind the desire for fierce competition in economy, satisfaction of customer’s demands, addressing challenges faced in the industries to enable production of right quantity and quality of goods in the most economic time and at minimum cost. Industrial Engineering plays a pivotal role in meeting these challenges, various techniques in Industrial Engineering to analyze, improve work methods, eliminate waste, proper allocation and utilization of appropriate resources.

HISTORY OF INDUSTRIAL ENGINEERING

The nature of industrial engineering (IE) as a profession will be more appreciated when its historical development is reviewed. For the purposes of this course, a brief history will be captured.

It has been suggested that perhaps Leonardo da Vinci was the first Industrial Engineer, due to the evidence showing his application of science to the analysis of human work, examining the rate at which a man could shovel dirt around the year 1500. Others state that the IE profession grew from Charles Babbage’s study of factory operations and specifically his work on the manufacture of straight pins in 1832. However, it has been generally argued that these early efforts were merely observational and did not attempt to engineered the jobs studied or increase overall output.

Most analysts agree that the IE profession grew out of application of science to the design of work and production systems that started in the 1880. The pioneers who led that effort were Frederick W. Taylor, who initiated the field of work measurement; Frank Gilbreth and his wife, Lillian Gilbreth (immortalized in the

1948 book and 1950 movie 'Cheaper by the Dozen'), who perfected methods improvement; and Henry Gantt pioneered the field of project management.

The term "Industrial Engineering" can therefore be traced to James Gunn, in The Engineering Magazine in 1901 when he suggested that a profession and curriculum of IE be established and organized similarly to that of electrical and mechanical engineering. With the first course in Industrial Engineering traced to the then 'Factory Economics' offered in the Mechanical Engineering department at the University of Kansas in the 1901-1902. While the first formal IE degree awarding program was established in 1908-1909 by Hugo Deimer at Penn State University as a modified version of the mechanical engineering curriculum.

The American Institute of Industrial Engineers, now known as the Institute of Industrial and Systems Engineers (IISE), was founded in 1948. Today there are over 100 IE degree awarding programs in the US alone.

Today, IE has and is one of the backbone of the developed countries, while became the most sought, and the big dreams of the developing countries if can be achieved. The general mild stones faced by the under-developed and developing countries today is mostly the IE related.

Contributions of IE in Industries

Some notable contributors in IE can be credited to some industrialist such as:

- Adam Smith (1776) through his book titled 'Wealth of Nations', laid foundation to scientific manufacturing. His concept was able to influence the division of labour which included the skill development, time savings and the use of specialized machine.
- James Watt (1864) utilizes 'Steam Engine' in advancing the use of mechanical power to increase productivity.
- Frederick Taylor (1859-1915) was a mechanical engineer and initiated investigations of better work methods and develop an integrated theory of management principles and methodologies. He proposed the following models:
 - Data Collection and standards for workers.
 - Scientifically training of workers.
 - Cooperation between management and labour for better production.
 - Divide work between management and labour and assigning to those who are best suited.
- Henry L. Gantt (1993) contributions includes;
 - Work in the area of motivation field, development of task and bonus plan.
 - Measurement of management results by Gantt Charts.
 - Recognition of social responsibility of business and industry.
 - Advocated training of workers by management.

- Frank and Lillian Gilbreth (1917) developed method study as a tool for work analysis. He developed micro-motion study, a breakdown of work into fundamental elements called 'therbligs'.
- Harrington Emerson (1913) developed his managerial concepts simultaneously with Taylor, Gantt and Gilbreth. Amongst his contributions is the Emerson's efficiency, bonus Plan, and the proposal of the twelve principles of efficiency.
- L.H.C Tippet (1937): Developed the concept of work sampling to determine the equipment and manpower utilization and setting performance standards for long cycle, heterogeneous jobs involving team work.

Industrial Engineering Approach

Some must effective and commonly experienced approaches of IE include the abilities in carrying out various activities by the industrial engineer; gathering and analyzing facts; preparing the alternative solutions taking into consideration all the constraints both internal and external; and selection of the best solution for implementation.

At problem identification/definition stage, the following steps must be taken:

- a) All the facts about the operation are collected and recorded using various recording techniques like charts, diagrams or models.
- b) Critical Examination of all facts by asking series of questions.
- c) Alternative ways are found by techniques like brainstorming.
- d) Based upon the criteria fixed for evaluation, the best alternative is selected.

Industrial Engineering Techniques

The following tools and techniques are mostly used to improve productivity of the organization by optimum utilization of resources: Method Study, Time Study (Work Measurement), motion economy, financial and non-financial incentives, value analysis, production, planning and control, inventory control, job evaluation, material handling analysis, ergonomics (human engineering), system analysis and the operations research techniques.

Approaches in IE

The new approaches utilized in IE today includes Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), Supply Chain Management (SCM), Business Intelligence (BI), Systems Dynamics (SD) and Software as a Service (SaaS).

Concepts of System Engineering

Systems Engineering is a transdisciplinary and integrative approach to enable the successful realization, use, and retirement of engineered systems, using systems principles and concepts that are inculcated in scientific, technological, and appropriate management methods. Thus, the terms “engineering” and “engineered” as used in the widest sense can mean “the action of working artfully to bring something about”; and “Engineered systems” may be composed of any or all of people, products, services, information, processes, and natural elements. These concepts can be summarized as follows:

- Establishing, balancing and integrating stakeholders’ goals, purpose and success criteria, and defining actual or anticipated customer needs, operational concept and required functionality, starting early in the development cycle.
- Establishing an appropriate lifecycle model, process approach and governance structures, considering the levels of complexity, uncertainty, change, and variety.
- Generating and evaluating alternative solution concepts and architectures.
- Baselineing and modelling requirements and selected solution architecture for each phase of the endeavor.
- Ware forming design synthesis and system verification and validation.
- While considering both the problem and solution domains, taking into account necessary enabling systems and services, identifying the role that the parts and the relationships between the parts play with respect to the overall behaviour and performance of the system, and determining how to balance all of these factors to achieve a satisfactory outcome.

Work System Design Issues

The concept of the work system in human factors and ergonomics (HF&E) is about the concept of work system design, criteria of work system design and the strategies of work system design

The Concept of Work System Design: In HF&E, the work system comprises workers and work equipment acting together to perform the system function in a convenient workspace, work environment and under the conditions imposed by the work tasks.

The Criteria of Work System Design: Design in HF&E strives to facilitate; feasibility of work, freedom from harm, freedom from physical and mental impairment and development of learning, health and personality.

The Strategies of Work System Design: HF&E design strategies are related to humans interacting within work systems (i.e. design to fit the whole working population or design for specific groups of users), to the interaction process (i.e. design for the work system task or design for specific requirements) and to the technical and environmental components (i.e. design of new work system or corrections of a work system).

An Example of a Model Based Control Development System

